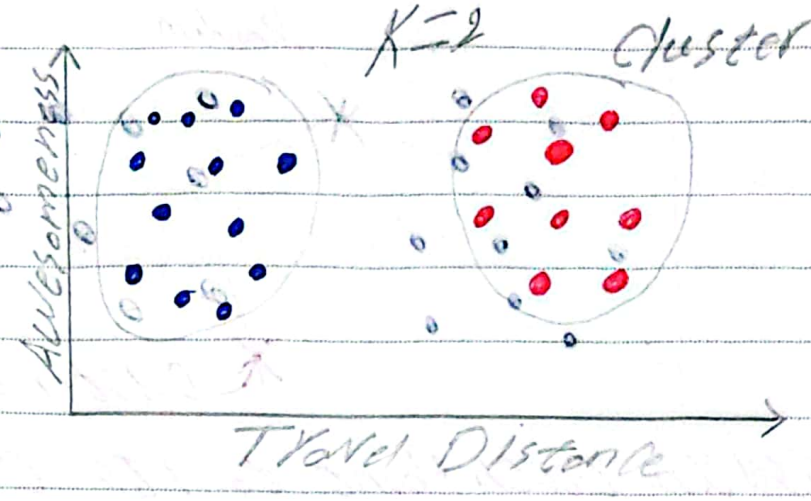
التاريخ: / / الموضوع:

Unsupervised

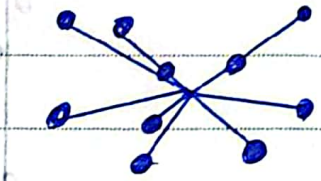
Clustering:



→ Number of K

Elbow Method:

K=1 avg distance = 1.261

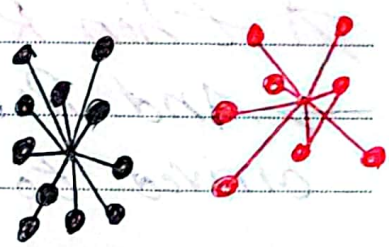


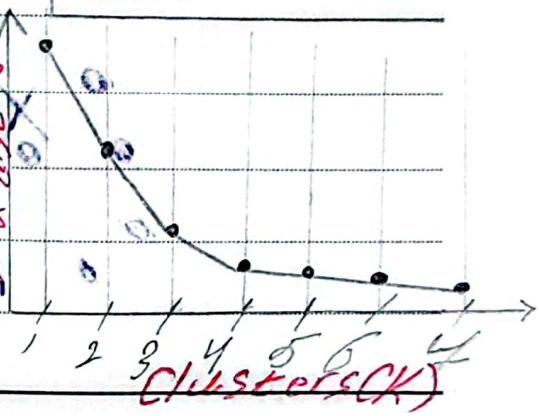
K=2 avg dist = .923

K=3 avg dist = .639

K=4 avg dist = .382

K=5 avg dist = .348



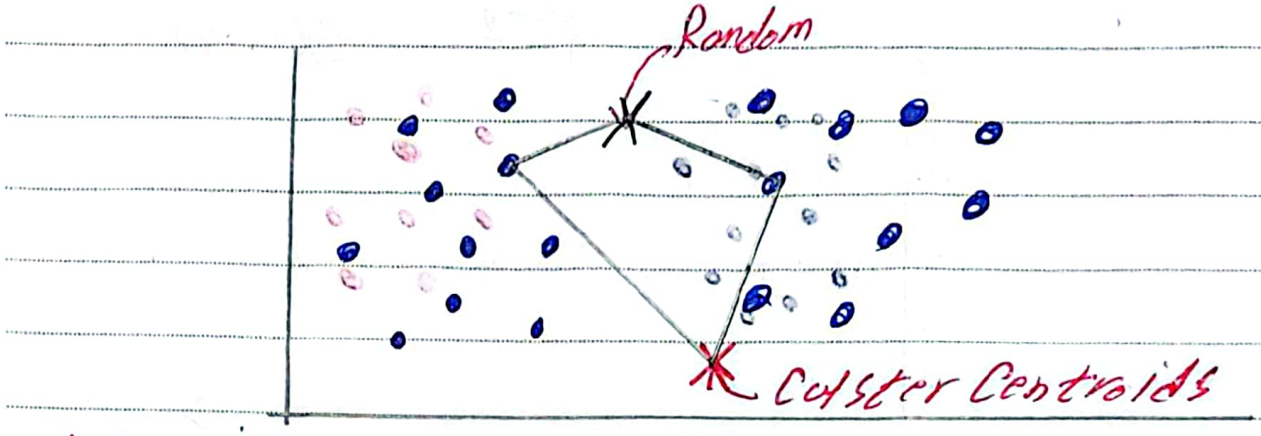


→ Large decreases at small K

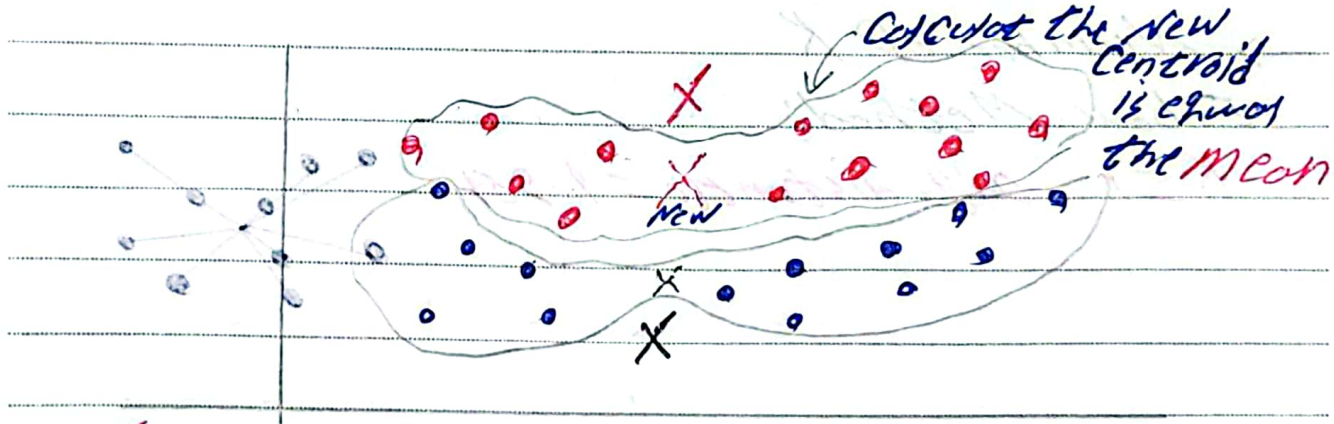
→ Small decreass at large K

→ Best Choice around elbow of Curve

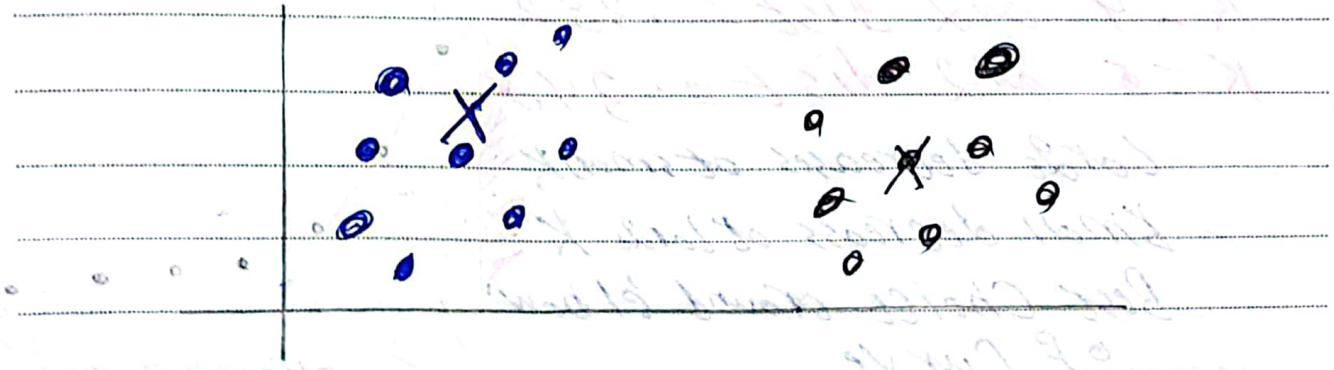
① After select The number of K'
② Randomly initialize Centroids:



③ Assign each Point to its Closest Centroid



④ then Recompute the Centroids
→ And Repeated two Action to arrived to no Point Change his Color



الموضوع: _____ التاريخ: / /

Feature Scaling

→ Transform numerical features to small scale

→ Types of feature scaling

[1] Standardization (Z-score scaling)

Transform data so it has:

Mean = 0

Standard deviation = 1

$$Z = \frac{X - \mu}{\sigma}$$

used in many algorithms like:

- SVM , Logistic Regression

[2] Min-Max Scaling (Normalization)

Transform values to a fixed range [0, 1]

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

→ Min-Max scaling (bad with outliers)

→ Robust scaling (best with outliers)

$$X' = \frac{X - Median}{IQR}$$

الموضوع: _____ التاريخ: / /

Note to Avoid Data Leakage

❌ Wrong Way scaling before splitting

```
scaler = standardscaler()
X_scaled = scaler.fit_transform(X)  # use of data
```

✅ Correct Way

```
X_train, X_test, y_train, y_test
    = train_test_split(X, y)
scaler = standardscaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Best Practice [Pipeline]

```
Pip = Pipeline([
        ("scaler", standardscaler()),
        ("model", LogisticRegression())
    ])
Pip.fit(X_train, y_train)
```

الموضوع: _____ التاريخ: / /

* When use:

KNN, K-Means, SVM, Logistic

Neural Networks

→ Not Require scaling:

Random Forest, Decision Tree